Biochemistry Cume November 18, 2016

The venom of the Wagler's pit viper contains a peptide that exhibits strong muscle relaxing effect. The peptide is also found in certain anti-aging cream. The primary amino-acid sequence is shown below:

Ser-Leu-Gly-Gly-Lys-Pro-Asp-Leu-Arg-Pro-Cys-Lys-Pro-Pro-Cys-His-Tyr-Ile-Pro-Arg-Pro-Lys-Pro-Arg

A. [10pts] Write out the 1 letter code for the above peptide.

B. [20pts] Calculate the net charge of the peptide at pH 2 and pH 7.

C. [30pts] Write out the products you would generate if you treated a portion of the peptide separately with trypsin, chymotrypsin and cyanogen bromide. Write out the products for each. Use the one letter code.

D. [20pts] The chymotrypsin treated toxin was applied on a carboxymethyl cellulose column (pKa=4). Write out the order of emergence of peptide fragments from the column when a linear salt gradient (buffered at pH=7) was applied.

E. [20pts] When treated with trypsin, the protein migrates on polyacrylamide gels under non-reducing conditions with the same molecular weight as the native protein. However, prior denaturation with urea + beta-mercaptoethanol, followed by removal of the beta-mercaptoethanol by dialysis, produces lower molecular weight peptide upon trypsin treatment. What does this information tell you?